

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284084

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Turner; Daniel R.

METHOD AND APPARATUS FOR DEPLOYMENT OF A COMMUNICATION LINE ONTO A SURFACE

Abstract

The subject matter of this specification can be embodied in, among other things, a method for adhering a tubular assembly onto a constructed surface, and the method includes removing a portion of the constructed surface to form a recess below a major plane of the constructed surface, arranging the tubular assembly partly within the recess, applying an uncured protectant, and curing the uncured protectant to protectively encase the tubular assembly.

Inventors: Turner; Daniel R. (Gainesville, VA)

Applicant: TRAXyL, Inc. (Gainesville, VA)

Family ID: 96948984

Appl. No.: 19/070250

Filed: March 04, 2025

Related U.S. Application Data

us-provisional-application US 63561578 20240305

Publication Classification

Int. Cl.: G02B6/50 (20060101)

U.S. Cl.:

CPC G02B6/504 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Application No. 63/561,578, filed on Mar. 5, 2024, the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to systems, methods, and apparatuses laying out a communication line and using a protectant to adhere and encase the communication line onto a surface.

BACKGROUND

[0003] Communication lines or communication cables (i.e., examples of tubular bodies) provide useful conduits to transfer information (e.g., audio, visual, etc.) using modern technology. An example of a communication line is a fiber optic cable. The deployment and installation of communication lines, however, can be challenging for several reasons. For example, communication lines may need to extend over relatively long distances to, e.g., provide telecommunications to a commercial or residential building. Additionally, the communication lines may need to be accessible (e.g., for maintenance or repairs). At the same time, however, the communication lines may need to be protected from the environment and are often preferred to be hidden from view.

[0004] The usual process for installing a communication cable over a relatively long distance may include extruding plastic core tubes around, e.g., fibers, helically wrapping the tubes around a central strength member, building up several layers of protective sheathing, surrounding the sheathing with a protective material (armor) such as steel, and surrounding the armor with a heavy polymer jacket. Once a cable is manufactured, it can be pulled through conduits, hung along telephone poles, and/or buried inside trenches. Additional possibilities include what is known as “micro trenching” (see e.g., International Patent Application Publication No. 09/935,346), where a small trench is cut into the road surface or the ground. The cable is then installed within the micro-trench or installed on a roadway in the layers of protective sheathing as described above. Another example is described in U.S. Patent Application Publication No. 2002/0038716 where a flat fiber conduit is used for further protection.

[0005] The use of armor and other materials can increase the weight, fabrication cost, and installation footprint of the communication cable. The larger installation footprint often requires significant disruption and repair of the installed route, each of which can contribute to higher installation cost.

[0006] When installing below the surface, communication cables can either be directly buried or installed into a conduit to further protect the cable. As noted above, the cable and conduit are often installed by trenching into the ground. The use of the conduit requires the added cost and overhead of conduit installation along with the fiber cable manufacturing and installation. Additionally, cutting the micro-trench adds expense and time to the installation process.

[0007] Plowing, direct burial, or other similar cable installation methods often require an array of machines. For example, a machine that utilizes a cutting blade to trench through soil or other natural ground substances may be needed. The communication cable may be installed in a process described in French Patent No. 2,762,155, which involves: (i) cutting a trench, (ii) burying a cable in the trench; and then (iii) backfilling the trench with a filler material suitable for the ground surface. This installation technique may cause significant disruption to local roadway infrastructure with attendant business impacts and transportation delays.

[0008] Installing a communication cable by micro-trenching is similar to a trenching or plowing install, but the cutting tool and ground opening are smaller. Less filler material may also be required. The micro-trenching installation technique is mainly used on roadways and often utilizes a heavy duty diamond saw blade that cuts a narrow (e.g., only a few inches wide) section of a roadway or similar surface to a predetermined depth depending on the location. A communication

cable is installed, or micro-duct is inserted for later filling with a communication cable. After that, the micro-trench is backfilled with an aggregate and another acceptable material. The risks of this technique (e.g., as described in International Patent Application Publication No. 09/935,346) include accidentally severing an existing utility line, damaging the integrity of the road surface, and risks to installers due to traffic and other safety concerns. Micro-trenching is not preferred in surfaces such as in private roads, shallow road surfaces, bridges, and tunnels, to name a few. [0009] The present disclosure is directed to overcoming these and other problems of the prior art and may provide a more efficient and/or more robust technique to deploy a tubular assembly, such as a communication line, in or on a paved surface.

SUMMARY

[0010] In general, this document systems, methods, and apparatuses laying out a communication line and using a protectant to adhere and encase the communication line onto a surface.

[0011] In an example implementation, a method for adhering a tubular assembly onto a constructed surface includes removing a portion of the constructed surface to form a recess below a major plane of the constructed surface, arranging the tubular assembly partly within the recess, applying an uncured protectant, and curing the uncured protectant to protectively encase the tubular assembly.

[0012] Various implementations can include some, all, or none of the following features. The method can include shaping the uncured protectant before curing the uncured protectant such that the uncured protectant has a different cross-sectional shape after the shaping than prior to the shaping. The shaping can be performed after the uncured protectant is applied onto the tubular assembly. The tubular assembly can be applied directly to the recess, and the uncured protectant is applied directly onto the tubular assembly. The curing of the uncured protectant can form a cured protectant that protectively encases the tubular assembly at least partly within the recess, the cured protectant being an outermost layer protecting the tubular assembly at least partly within the recess. The tubular assembly can have a predetermined thickness, and the recess has a depth less than the predetermined thickness. The method can include applying the uncured protectant onto the tubular assembly, and curing the uncured protectant to cover the tubular assembly. The method can include applying the uncured protectant within the recess, and curing the uncured protectant to cover the tubular assembly. A first portion of the tubular assembly can be arranged below the major plane and a second portion of the tubular assembly can extend beyond the major plane, and the uncured protectant can be cured to protectively encase the first portion within the recess and cover the second portion beyond the major plane. The tubular assembly can include an optical conductor.

[0013] In an example embodiment, an apparatus for adhering a tubular assembly to a constructed surface includes a main body moveable in an advancing direction, the main body possessing a forward end and a rear end, a receiver configured to store a tubular assembly having a predetermined thickness, the receiver being mounted on the main body, a container configured to store an uncured protectant, the container being mounted on the main body, a surface modification apparatus connected to the main body, the surface modification apparatus being configured to contact the constructed surface at a contact point to form a recess below a major plane of the constructed surface when the apparatus moves in the advancing direction, a tubular assembly applicator configured to receive a portion of the tubular assembly from the receiver and configured to arrange the tubular assembly partly within the recess, and a protectant applicator connected to the container and configured to eject the uncured protectant wherein the uncured protectant is configured to cure after the uncured protectant is applied.

[0014] Various embodiments can include some, all, or none of the following features. The surface modification apparatus can be a grinder. The surface modification apparatus can be a wheel. The uncured protectant can be a resin. The uncured protectant can be configured to be cured by application of a catalyst configured to initiate a curing process. The surface modification apparatus can be configured to form the recess to a depth less than the predetermined thickness below the major plane. The protectant applicator can be configured to eject the uncured protectant onto the

tubular assembly, and the protectant applicator is positioned such that ejected uncured protectant is applied after the tubular assembly has been arranged by the tubular assembly applicator. The protectant applicator can be configured to eject the uncured protectant into the recess, and the protectant applicator is positioned such that ejected uncured protectant is applied after the tubular assembly has been arranged by the tubular assembly applicator. The protectant applicator can be configured to eject the uncured protectant into the recess, and the protectant applicator is positioned such that ejected uncured protectant is applied before the tubular assembly is arranged by the tubular assembly applicator. The tubular assembly applicator can be configured to arrange the tubular assembly partly within the recess such that a first portion of the tubular assembly is arranged below the major plane and a second portion of the tubular assembly extends beyond the major plane. The tubular assembly can include an optical conductor.

[0015] In another example embodiment, a tubular assembly includes a tubular body having a predetermined first cross-sectional shape and a predetermined first cross-sectional area, and a compliant jacket coaxially surrounding the tubular body and having a predetermined second cross-sectional shape and a predetermined second cross-sectional area.

[0016] Various embodiments can include some, all, or none of the following features. The tubular body can include one or more elements comprising one or more of an optical conductor, an electrical conductor, a thermal conductor, or luminal tube. The tubular body can include two or more of the elements arranged with a predetermined spacing. The tubular assembly can include a protective covering coaxially surrounding the tubular body. The compliant jacket can include an at least partly compressible material, wherein the compliant jacket is at least partly compressible from the predetermined second cross-sectional shape to a third cross-sectional shape different from the second cross-sectional shape and having a third cross-sectional area less than the second cross-sectional area. The compliant jacket can include a substantially incompressible material, wherein the compliant jacket is transformable from the predetermined second cross-sectional shape to a third cross-sectional shape different from the second cross-sectional shape and having the second cross-sectional area. The compliant jacket can include an elastic material, wherein the compliant jacket is configured to be distorted from the predetermined second cross-sectional shape to a third cross-sectional shape different from the second cross-sectional shape and then at least partly recover the second cross-sectional shape. The compliant jacket can include a material that is curable to substantially maintain the second cross-sectional shape or a third cross-sectional shape different from the second cross-sectional shape.

[0017] In another example implementation, a method for adhering a tubular assembly onto a constructed surface includes receiving the tubular assembly having a tubular body having a predetermined first cross-sectional shape and a predetermined first cross-sectional area, and a compliant jacket coaxially surrounding the tubular body and having a predetermined second cross-sectional shape and a predetermined second cross-sectional area, arranging the tubular assembly partly within a recess having a third cross-sectional shape defined in a major plane of the constructed surface and a third cross-sectional area, wherein a sum of the first cross-sectional shape and the second cross-sectional area is substantially equal to or less than the third cross-sectional area, and at least partly conforming the compliant jacket from the second cross-sectional shape to the third cross-sectional shape.

[0018] Various implementations can include some, all, or none of the following features. The method can include modifying the second cross-sectional shape to a fourth cross-sectional shape, different from the second cross-sectional shape, and at least partly conforming the compliant jacket from the fourth cross-sectional shape to the third cross-sectional shape. Modifying the second cross-sectional shape to a fourth cross-sectional shape, different from the second cross-sectional shape can include compressing the compliant jacket such that the second cross-sectional area to a fourth cross-sectional area less than the second cross-sectional area. At least partly conforming the compliant jacket from the fourth cross-sectional shape to the third cross-sectional shape can include

expanding the compliant jacket from the fourth cross-sectional shape to a fifth cross-sectional area larger than the fourth cross-sectional shape. The compliant jacket can include a substantially incompressible material, and the fourth cross-sectional shape has substantially the second cross-sectional area. The tubular body can be arranged below the major plane. The method can include modifying a portion of the constructed surface to form the recess below the major plane of the constructed surface. At least partly conforming the compliant jacket from the second cross-sectional shape to the third cross-sectional shape can include applying a compressive force over the tubular assembly. Applying a compressive force over the tubular assembly can include rolling a wheel over the tubular assembly and the recess.

[0019] In another example embodiment, an apparatus for adhering a tubular assembly to a constructed surface includes a main body moveable in an advancing direction, the main body possessing a forward end and a rear end, a receiver configured to store a tubular assembly, the receiver being mounted on the main body and the tubular assembly having a tubular assembly having a predetermined first cross-sectional shape and a predetermined first cross-sectional area, and a compliant jacket coaxially surrounding the tubular assembly and having a predetermined second cross-sectional shape and a predetermined second cross-sectional area, a tubular assembly applicator configured to receive a portion of the tubular assembly from the receiver and configured to arrange the tubular assembly at least partly within a recess having a third cross-sectional shape defined in a major plane of the constructed surface and a third cross-sectional area, wherein a sum of a first cross-sectional area of the first cross-sectional shape and second cross-sectional area of the second cross-sectional shape substantially equals the third cross-sectional area, and a compressor configured to apply a compressive force over the tubular assembly to at least partly conform the compliant jacket from the second cross-sectional shape to the third cross-sectional shape.

[0020] Various embodiments can include some, all, or none of the following features. The apparatus can include a surface modification apparatus connected to the main body, the surface modification apparatus being configured to contact the constructed surface at a contact point to form the recess below a major plane of the constructed surface. The apparatus can include a finisher configured to cure the compliant jacket to the third cross-sectional shape.

[0021] The systems and techniques described here may provide one or more of the following advantages. First, a system can provide communications across long distances. Second, the system can reduce the cost of deploying communications cables. Third, the system can increase the speed of communication cable deployment. Fourth, the system can reduce the amount of modification performed to deployment surfaces and locations across which a communication cable is being deployed.

[0022] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0023] FIG. 1A show an example of a tubular assembly embedded in a constructed surface.

[0024] FIG. 1B show another example of a tubular assembly embedded in a constructed surface.

[0025] FIGS. 2A-2E show an example of embedding a tubular assembly in a constructed surface.

[0026] FIGS. 3A-3D show another example of embedding a tubular assembly in a constructed surface.

[0027] FIGS. 4A-4C show another example of embedding a tubular assembly in a constructed surface.

[0028] FIGS. 5A-5C show another example of embedding a tubular assembly in a constructed surface.

[0029] FIGS. 6A-6C show another example of embedding a tubular assembly in a constructed surface.

[0030] FIG. 7 is a flow diagram of an example process for embedding a tubular assembly in a constructed surface.

[0031] FIG. 8 is a flow diagram of another example process for embedding a tubular assembly in a constructed surface.

[0032] FIG. 9 shows a perspective view of a tubular assembly deployment apparatus according to embodiments of the disclosure.

[0033] FIG. 10 shows a roller machine equipped with a tubular deployment apparatus according to embodiments of the disclosure.

[0034] FIG. 11 shows an asphalt paver machine equipped with a tubular deployment apparatus according to embodiments of the disclosure.

DETAILED DESCRIPTION

[0035] This document describes systems and techniques for systems, methods, and apparatuses for laying out a communication line or other tubular assembly and using a protectant to adhere and encase the communication line onto a surface. In general, a constructed surface (e.g., pavement, wall) is modified to form a recess having a predetermined shape, and a communication or other tubular assembly having a shapable outer body is deployed within the recess. The shapable outer body is configured to substantially conform to the predetermined shape of the recess such that the tubular assembly becomes substantially retained within the recess.

[0036] FIG. 1A illustrates an example tubular assembly **150a** embedded in a constructed surface **100a** such as a road, runway, sidewalk, floor, wall, ceiling, or any other appropriate surface. As indicated by the dotted lines in FIG. 1A, the tubular assembly **150a** is underneath a top surface **102a** of the constructed surface **100a**. The tubular assembly **150a** can be deployed towards the edge of the surface **100a**. For example, the tubular assembly **150a** can be embedded near the shoulder of a road. This can beneficially decrease forces exerted on the tubular assembly **150a** by vehicles traveling on the surface **100a**. In other embodiments, the tubular assembly **150a** can be deployed in the center of a road (e.g., beneath the painted road markings) or in any other appropriate location of the road. In some embodiments, the tubular assembly **150a** can be arranged to extend generally longitudinally in parallel to a direction of traffic (e.g., to reduce the frequency of cars or other surface loads crossing over the tubular assembly **150a**).

[0037] In the illustrated example, the tubular assembly **150a** is partly arranged or embedded within a recess **105a** defined in the top surface **102a** and is then covered by a protective layer **110a**. In some embodiments, the tubular assembly **150a** may be partly within and partly above the recess **105a**. In some embodiments, a fiber optic cable of the tubular assembly **150a** may be partly within and partly above the recess **105a**, and in other embodiments the fiber optic cable of the tubular assembly **150a** may be fully within the recess **105a** and remaining parts of the tubular assembly **150a** (e.g., protective sheathing) may extend above the recess **105a**. The protective layer **110a** provides protection for the tubular assembly **150a**, and also provides a smooth surface over a portion of the tubular assembly that extends above the top surface **102a**. An example of this configuration will be discussed further in the descriptions of FIGS. 2A-2D.

[0038] FIG. 1B illustrates an example tubular assembly **150b** embedded in a constructed surface **100b** such as a road, runway, sidewalk, floor, wall, ceiling, or any other appropriate surface. For example, the tubular assembly **150b** can be embedded near the shoulder of a road. This can beneficially decrease forces exerted on the tubular assembly **150b** by vehicles traveling on the constructed surface **100b**. In other embodiments, the tubular assembly **150b** can be deployed in the center of a road (e.g., beneath the painted road markings) or in any other appropriate location of the road. In some embodiments, the tubular assembly **150b** can be arranged to extend generally

longitudinally in parallel to a direction of traffic (e.g., to reduce the frequency of cars or other surface loads crossing over the tubular assembly **150b**).

[0039] In the illustrated example, the tubular assembly **150b** is at least partly arranged or embedded within a recess **105b** defined in the top surface **102b**. As will be discussed in more detail in the descriptions of FIGS. 3A-6C, the recess **105b** is configured to mechanically retain the tubular assembly **150b**, and the tubular assembly **150b** is configured to at least partly conform to the recess **105b** to resist disengagement from the constructed surface **100b**.

[0040] Paved surfaces can include many layers. In the embodiment illustrated in FIG. 2, the paved surface is a road that includes a base layer **101a**. The base layer **101a** can include one or more layers of an asphalt layer and/or a concrete layer.

[0041] In some embodiments, the tubular assembly **150a** and/or **150b** can be embedded within any layer of a road and/or in-between any layers of the road during the road building process. In some embodiments, the constructed surface materials can include gravel, earth, asphalt, concrete, or combinations or layers of these and/or any other appropriate materials. In some embodiments, the tubular assembly **150a** and/or **150b** can be embedded within any layer of a structure and/or in-between any layers of the structure during a structural building process. In some embodiments, the constructed structural materials can include drywall, wood, engineered wood, fiberglass, plaster, plastic, cut or engineered stone slabs, high density foam (e.g., Styrofoam), or combinations or layers of these and/or any other appropriate materials.

[0042] FIGS. 2A-2E show an example of embedding a tubular assembly **250** in a constructed surface **201**. In some embodiments, the tubular assembly **250** can be the example tubular assembly **150a** or **150b** of FIGS. 1A and 1B. In some embodiments, the constructed surface **201** can be the example constructed surface **101a** or **101b** of FIGS. 1A and 1B.

[0043] FIGS. 2A and 2B show an example tool **270** being used to contact the top surface **202** at a contact point and define a recess **280** below a top surface **202** of the constructed surface **201**. In some embodiments, the tool **270** can be any appropriate surface modification apparatus that can be used to remove a portion of the constructed surface **201** to define the recess **280** (e.g., a trench, groove, depression) with a substantially predetermined shape. For example, the tool **270** can be a saw, a grinder, or a router configured to rotate or reciprocate to cut or loosen the material of the constructed surface **201** to form the recess **280**. In another example, the tool **270** can be a wheel configured to apply compressive force against the top surface **202** and compact, compress, or displace the material of the constructed surface **201** to form the recess **280**. In another example, the tool **270** can be a blade configured to cut the top surface **202** and displace the material of the constructed surface **201** to form the recess **280**.

[0044] FIG. 2C shows the tubular assembly **250** being arranged such that it is partly retained within the recess **280**. The tubular assembly **250** includes a tubular body **252** coaxially surrounded by protective jacket **254**.

[0045] In some embodiments, the tubular body **252** can have a predetermined thickness, a predetermined cross-sectional shape, and/or a predetermined cross-sectional area (e.g., the tubular body **252** can have a substantially fixed cross-sectional shape that is substantially incompressible). In some embodiments, the tubular body can include one or more elements comprising one or more of an optical conductor, an electrical conductor, a thermal conductor, or luminal tube. For example, the tubular body **252** can be bare or clad optical fiber.

[0046] In some embodiments, the protective jacket **254** can have a predetermined cross-sectional shape and/or a predetermined cross-sectional area (e.g., the protective jacket **254** can have a substantially fixed cross-sectional shape that is substantially incompressible). In some embodiments, the protective jacket **254** can be a compliant jacket that made of an at least partly compressible material and/or deformable material. Examples of compliant jackets will be discussed in more detail in the descriptions of FIGS. 3A-6C.

[0047] In the illustrated example, the recess **280** is defined with a width that is complimentary to

(e.g., approximately equal to or slightly wider than) the width of the tubular assembly **250** and is defined with a depth that is less than the height of the tubular assembly **250**. In the illustrated example, the recess **280** has a depth that is slightly greater than the radius of the tubular assembly **250**.

[0048] Referring to FIG. 2D, the tubular assembly **250** has been laid down at least partly within the recess **280**. As best shown in this view, the recess **280** has a depth such that the tubular body **252** is partly or entirely arranged below the plane of the top surface **202**.

[0049] FIG. 2E shows the constructed surface **201** after the tubular assembly **250** has been encased within the recess **280** by a protectant **210** (e.g., resin). In general, after the tubular assembly **250** has been arranged within the recess **280**, an uncured protectant is applied (e.g., poured, brushed, sprayed) over the tubular assembly **250** and the recess **280**. The uncured protectant is configured to cure after the uncured protectant is applied to form the protectant **210** to protectively encase tubular assembly **250** and adhere or otherwise retain the tubular assembly **250** to the constructed surface **201**. For example, the protectant **210** can provide physical protection against actions on the top surface **202** (e.g., wheels, foot traffic, flowing water or other fluids, abrasion from sand or other debris). In another example, the protectant **210** can provide physical protection against ambient conditions on the top surface **202** (e.g., the sun's ultraviolet rays, road salt or deicers, cleaners or other chemicals, ozone). In another example, the protectant **210** can substantially seal the recess **280** against ingress of materials from the top surface (e.g., to prevent water from entering the recess **280** and freezing). In another example, the protectant **210** can provide a smooth surface for surface traffic (e.g., to make it easier for wheels to traverse the recess **280**, to reduce tripping hazards, to reduce turbulence in flowing fluids). For example, the uncured protectant can be or include a resin. The uncured protectant may be, for example, a methyl methacrylate (MMA) that can be cured by being mixed with a catalyst. For example, benzoyl peroxide (BPO) can be used as a catalyst to interact with MMA. Additionally or alternatively, the uncured protectant may include an epoxy and/or polyurea. For example, a two-part epoxy may be used that mix immediately prior to, during, or immediately after being applied. Epoxy or polyurea primers may also be used. Protective flakes may be used or added to the uncured protectant. These materials are simply discussed to provide some examples. Any combinations of these materials and other appropriate known protective materials may be utilized for the uncured protectant.

[0050] FIGS. 3A-3D show an example of embedding an example tubular assembly **350** in an example constructed surface **301**. In some embodiments, the tubular assembly **350** can be the example tubular assembly **150a** or **150b** of FIGS. 1A and 1B. In some embodiments, the constructed surface **301** can be the example constructed surface **101a** or **101b** of FIGS. 1A and 1B.

[0051] FIGS. 3A and 3B show an example tool **370** being used to define a recess **380** below a top surface **302** of the constructed surface **301**. In some embodiments, the tool **370** can be any appropriate surface modification apparatus that can be used to remove a portion of the constructed surface **301** to define the recess **380** (e.g., a trench, groove, depression) with a substantially predetermined shape. For example, the tool **370** can be a saw or a router configured to rotate or reciprocate to cut or loosen the material of the constructed surface **301** to form the recess **380**. In another example, the tool **370** can be a wheel configured to apply compressive force against the top surface **302** and compact, compress, or displace the material of the constructed surface **301** to form the recess **380**. In another example, the tool **370** can be a blade configured to cut the top surface **302** and displace the material of the constructed surface **301** to form the recess **380**.

[0052] FIG. 3C shows the tubular assembly **350** being arranged such that it is partly or entirely retained within the recess **380**. The tubular assembly **350** includes a tubular body **352** coaxially surrounded by protective jacket **354**.

[0053] In some embodiments, the tubular body **352** can have a predetermined cross-sectional shape and/or a predetermined cross-sectional area (e.g., the tubular body **352** can have a substantially fixed cross-sectional shape that is substantially incompressible). In some embodiments, the tubular

body can include one or more elements comprising one or more of an optical conductor, an electrical conductor, a thermal conductor, or luminal tube. For example, the tubular body **352** can be bare or clad optical fiber.

[0054] In some embodiments, the protective jacket **354** can have a predetermined cross-sectional shape and/or a predetermined cross-sectional area. For example, the protective jacket **354** can be flexible but substantially incompressible (e.g., elastic material, rubbery material) such that the shape of the protective jacket **354** can be distorted by external forces to permit passage into the recess **380**, and when those forces are relieved the protective jacket **354** can return toward its original shape within the recess **380**.

[0055] In some embodiments, the protective jacket **354** can be a compliant jacket that is compressible and/or deformable. For example, the protective jacket **354** can be partly made of an elastic foam or other material that can be compressed by external forces (e.g., squeezed, squished, compacted) to permit passage into the recess **380**, and when those forces are relieved the protective jacket **354** can re-expand toward its original shape within the recess **380**. Additional examples of compliant jackets will be discussed in more detail in the descriptions of FIGS. **4A-6C**.

[0056] Referring to FIG. **3D**, the tubular assembly **350** has been laid down at least partly within the recess **380**. As best shown in this view, the recess **280** has a depth such that the tubular body **352** is partly or entirely arranged below the plane of the top surface **302**.

[0057] FIGS. **4A-4C** show another example of embedding the example tubular assembly **350** in the example constructed surface **301**. In FIG. **4A**, the tubular assembly **350** is positioned over an opening **480** of the recess **380**. The tubular assembly **350** is shown in a default (e.g., pre-installation) configuration, in which the protective jacket **354** coaxially surrounding the tubular body **352** is in a substantially undeformed (e.g., undistorted, uncompressed) default shape. The tubular body **352** has a predetermined thickness, a predetermined cross-sectional shape, and a predetermined cross-sectional area (e.g., an electrical or optical conductor having a substantially round and substantially incompressible and/or distortable shape). The protective jacket **354** is a compliant jacket coaxially surrounding the tubular body **352** and has a predetermined cross-sectional shape and a predetermined cross-sectional area.

[0058] The protective jacket **354** is made of one or more materials that can be compressed and/or reshaped (e.g., squeezed, squished) from the predetermined configuration under force to have a different cross-sectional shape and/or a smaller cross-sectional area, and then flex and/or expand back toward the undeformed configuration when released. The protective jacket **354** may be, for example, an elastomer, plastic, rubber, foam, or gel. Any appropriate material that can be compressed and/or reshaped as described herein may be utilized for the protective jacket **354**.

[0059] In FIG. **4B**, the protective jacket **354** has been compressed and distorted from its default shape such that it can be passed through the opening **480** into the recess **380**. The example recess **380** is formed with a trapezoidal cross-sectional shape, in which the opening **480** defines what would otherwise be the minor base of the trapezoidal shape, and then the recess widens away from the opening **480** below the top surface **302**. In the illustrated example, this process of squeezing the tubular assembly **350** into the recess **380** continues until the tubular assembly **350** is substantially entirely within the recess **380** below the top surface **302**.

[0060] In FIG. **4C**, the tubular assembly **350** has been substantially entirely inserted within the recess **380** below the top surface **302**. Pressure has been applied to cause the protective jacket **354** to cause the protective jacket **354** to partly flow until it touches the walls of the recess **380** and substantially fill the available space within the recess **380**. The deformed shape of the protective jacket **354** at least partly retains the tubular assembly **350** and the tubular body **352** within the recess, substantially below the top surface **302**. In some embodiments, a portion of the protective jacket **354** may extend at least partly above the plane of the top surface **302**, and in some such examples, a curable protectant may be applied over the top of the tubular assembly **350** and the recess **380** (e.g., similar to the example protectant **210** of FIG. **2E**).

[0061] FIGS. 5A-5C show another example of embedding an example tubular assembly **550** in an example constructed surface **501**. In some embodiments, the tubular assembly **550** can be the example tubular assembly **150a** or **150b** of FIGS. 1A and 1B. In some embodiments, the constructed surface **501** can be the example constructed surface **101a** or **101b** of FIGS. 1A and 1B. [0062] In FIG. 5A, the tubular assembly **550** is positioned over an opening **581** of a recess **580**. The tubular assembly **550** is shown in a default (e.g., pre-installation) configuration, in which a protective jacket **554** surrounding an arrangement of tubular bodies **552** is in a substantially undeformed (e.g., undistorted, uncompressed) default shape that is configured to compliment the predetermined cross-sectional shape of the recess **580**. The tubular bodies **552** each have a predetermined cross-sectional shape and a predetermined cross-sectional area (e.g., an electrical or optical conductor having a substantially round and substantially incompressible and/or distortable shape).

[0063] The protective jacket **554** is a compliant jacket surrounding the arrangement of tubular bodies **552**. The tubular bodies **552** are spaced and arranged within the protective jacket **554** in a predetermined configuration. For example, the tubular bodies **552** can be optical fibers and can be spaced apart to substantially prevent optical crosstalk. In another example, the tubular bodies **552** can be electrical conductors and can be spaced apart to promote electromagnetic isolation, to promote electromagnetic shielding, and/or to maintain a predetermined inter-conductor impedance or capacitance. For example, the tubular assembly **550** can be configured as an Ethernet cable assembly in which twisted pairs of conductors are held in a predetermined physical arrangement with each other (e.g., as prescribed by IEEE 802.X specifications) by the protective jacket **554**.

[0064] The protective jacket **554** is made of one or more materials that can be distorted and/or reshaped (e.g., squeezed) from the predetermined configuration under force to have a different cross-sectional shape, and then flex back toward the undeformed configuration when released.

[0065] In FIG. 5B, the protective jacket **554** has been distorted from its default shape such that it can be passed through the opening **581** into the recess **580**. The example recess **580** is formed with generally rectangular cross-sectional shape, in which the opening **581** defines a narrow passage to a wider rectangular cavity below the top surface **502**. In the illustrated example, this process of squeezing the tubular assembly **550** into the recess **580** continues until the tubular assembly **550** is substantially entirely within the recess **580** below the top surface **502**.

[0066] In FIG. 5C, the tubular assembly **550** has been substantially entirely inserted within the recess **580** below the top surface **502**. Upon release, the protective jacket **554** self-restores, undistorting back toward its default shape such that it substantially fills the available space within the recess **580**, and returns the collection of tubular bodies **552** back to their predefined interpositional configuration. The partly restored shape of the protective jacket **554** at least partly retains the tubular assembly **550** and the tubular body **552** within the recess, substantially below the top surface **502**, and maintains the positional configuration of the collection of tubular bodies **552**. In some embodiments, a portion of the protective jacket **554** may restore at least partly above the plane of the top surface **502**, and in some such examples, a curable protectant may be applied over the top of the tubular assembly **550** and the recess **580** (e.g., similar to the example protectant **210** of FIG. 2E).

[0067] FIGS. 6A-6C show another example of embedding an example tubular assembly **650** in an example constructed surface **601**. In some embodiments, the tubular assembly **650** can be the example tubular assembly **150a** or **150b** of FIGS. 1A and 1B. In some embodiments, the constructed surface **601** can be the example constructed surface **101a** or **101b** of FIGS. 1A and 1B.

[0068] In FIG. 6A, the tubular assembly **650** is positioned over an opening **681** of a recess **680**. The tubular assembly **650** is shown in a default (e.g., pre-installation) configuration **651**, in which a protective jacket **654** coaxially surrounding a tubular body **652** is in a substantially undeformed (e.g., undistorted, uncompressed) default shape. The tubular body **652** has a predetermined cross-sectional shape and a predetermined cross-sectional area (e.g., an electrical or optical conductor

having a substantially round and substantially incompressible and/or distortable shape). The protective jacket **654** is a compressible jacket coaxially surrounding the tubular body **652** and has a predetermined cross-sectional shape and a predetermined cross-sectional area.

[0069] The protective jacket **654** is made of one or more materials that can be compressed and/or reshaped (e.g., squeezed, squished) from the predetermined configuration under force to a deformed configuration **653**, which has a different cross-sectional shape and/or a smaller cross-sectional area from the undeformed configuration **651**, and can fit through the opening **681** to be arranged within the recess **680**.

[0070] The example recess **680** is formed with a trapezoidal cross-sectional shape, in which the opening **681** defines what would otherwise be the minor base of the trapezoidal shape, and then the recess widens away from the opening **681** below the top surface **602**. While the example recess **680** is formed with a trapezoidal cross-sectional shape in the illustrated example, any appropriate cross-sectional shape can be used. For example, an inverted “T” shape (e.g., the example recess **580** of FIGS. 5A-5C), an ovoid or otherwise rounded shape could be used. In some embodiments, by defining the opening **681** to have a narrower width than at least a portion of the recess **680** below, mechanical retention of the tubular assembly **650** may be promoted. In the illustrated example, the process of squeezing the tubular assembly **650** into the recess **680** continues until the tubular assembly **650** is substantially entirely within the recess **680** below the top surface **602**.

[0071] In FIG. 6B, the tubular assembly **650** has been inserted into the recess **680** and the compression applied to the protective jacket **654** has been released. After release, the protective jacket **654** starts to expand away from the distorted configuration **653**.

[0072] In FIG. 6C, the protective jacket **354** has partly self-restored, at least partly undistorting and/or expanding back toward its default shape until it touches the walls of the recess **680** until it substantially fills the available space within the recess **680**. The partly restored shape of the protective jacket **654** at least partly retains the tubular assembly **650** and the tubular body **652** within the recess, substantially below the top surface **602**. In some embodiments, a portion of the protective jacket **654** may restore or re-expand at least partly above the plane of the top surface **602**, and in some such examples, a curable protectant may be applied over the top of the tubular assembly **650** and the recess **380** (e.g., similar to the example protectant **210** of FIG. 2E).

[0073] FIG. 7 is a flow diagram of an example process **700** for embedding a tubular assembly in a constructed surface.

[0074] At **710**, a portion of a constructed surface is removed to form a recess below a major plane of the constructed surface. For example, the example tool **270** of FIGS. 2A and 2B can be used to remove a portion of the constructed surface **201** to form the recess **280** below the top surface **202**.

[0075] At **720**, a tubular assembly is arranged partly within the recess. For example, as shown in FIGS. 2C and 2D, the example tubular assembly **250** can be at least partly inserted into the recess **280**.

[0076] At **730**, an uncured protectant is applied over the tubular assembly and the recess. For example, as shown in FIG. 2E, the example protectant **210** can be applied over the tubular assembly **250** in the recess **280**. In some embodiments, the protectant **210** can go on as a liquid or gel. In some embodiments, the protectant **210** can be a flexible solid in its uncured form (e.g., a tape or band) that can be laid down over the tubular assembly **250** and the recess **280**.

[0077] At **740**, the uncured protectant is cured to protectively encase the tubular assembly. For example, in some embodiments the protectant **210** can go on as a liquid or gel and then cured to a substantially solid state (e.g., evaporative curing, by exposing light sensitive protectant to ultraviolet light, a catalyst or other chemical curing agents can be applied). In some embodiments, the protectant **210** can be a flexible solid in its uncured form (e.g., a tape or band) that can be laid down over the tubular assembly **250** and the recess **280**, and then cured into a harder state.

[0078] In some embodiments, the uncured protectant can be shaped or formed before curing the uncured protectant such that the uncured protectant has a different cross-sectional shape after the

shaping than prior to the shaping. In some embodiments, the shaping can be performed after the uncured protectant is applied onto the tubular assembly. An example of a shaper for shaping uncured protectant will be discussed further in the description of FIG. 9.

[0079] In some embodiments, the tubular assembly can be applied directly to the recess, and the uncured protectant is applied directly onto the tubular assembly. For example, the example tubular assembly **250** can be inserted into the recess **280** without first distorting or compressing the tubular assembly **250**.

[0080] In some embodiments, the curing of the uncured protectant forms a cured protectant that protectively encases the tubular assembly at least partly within the recess, the cured protectant being an outermost layer protecting the tubular assembly at least partly within the recess. For example, the tubular body **252** can be arranged within a layer of the protectant **210** (e.g., with or without the protective jacket **254**), such that when the protectant **210** is cured, the protectant performs some or all of the functions of the protective jacket **254** (e.g., to act as an optical cladding or electrical insulator, to promote retention within the recess, to protect against hazards at the top surface **202**).

[0081] In some embodiments, the tubular assembly can have a predetermined diameter or thickness, and the recess can a depth that is less than the predetermined diameter or thickness. For example, the example recess **280** is not so deep that the tubular assembly **250** is completely recessed below the plane of the top surface **202**.

[0082] In some implementations, the process **700** can include applying the uncured protectant onto the tubular assembly and curing the uncured protectant to cover the tubular assembly. For example, in the examples shown in FIGS. 2A-2E, the protectant **210** covers the tubular assembly **250** and is cured to a substantially solid form.

[0083] In some implementations, the process **700** can include applying the uncured protectant within the recess and curing the uncured protectant to cover the tubular assembly. For example, the protectant **210** can be applied into the recess before the tubular assembly **250** is inserted. In examples such as this, the tubular assembly **250** can at least partly sink into the uncured protectant before the protectant is cured, retaining the tubular assembly **250** within the protectant.

[0084] In some implementations, a first portion of the tubular assembly can be arranged below the major plane and a second portion of the tubular assembly can extends beyond the major plane, and the uncured protectant can be cured to protectively encase the first portion within the recess and cover the second portion beyond the major plane. For example, the example tubular assembly **250** extends partly above the plane of the top surface **202**, and the protectant **210** not only encases the tubular assembly **250** within the recess, it also forms a protective cap over the top of the tubular assembly **250** and the recess **280**.

[0085] In some embodiments, the tubular assembly can include an optical conductor. For example, one or more of the example tubular assemblies **250**, **350**, **550**, and **650** can be fiber optic cables. In some embodiments, the tubular assembly can include an electrical conductor. For example, one or more of the example tubular assemblies **250**, **350**, **550**, and **650** can be electrical power or communication cables. In some embodiments, the tubular assembly can include a tubular lumen. For example, one or more of the example tubular assemblies **250**, **350**, **550**, and **650** can be a tube configured to carry a fluid.

[0086] FIG. 8 is a flow diagram of another example process **800** for embedding a tubular assembly in a constructed surface.

[0087] At **810** the tubular assembly is received. The tubular assembly includes a tubular body having a predetermined first cross-sectional shape and a predetermined first cross-sectional area, and a compliant jacket coaxially surrounding the tubular body and having a predetermined second cross-sectional shape and a predetermined second cross-sectional area. For example, in FIGS. 4A-4C, the example tubular assembly **350** includes the tubular body **352**, which is configured to substantially maintain its cross section shape, and the protective jacket **354**, which is configure to

be compressed and/or deformed away from an initial cross-sectional shape (e.g., as shown in FIG. 4A) to a second, different cross-sectional shape (e.g., as shown in FIGS. 4B and 4C).

[0088] At **820**, the tubular assembly is arranged partly within a recess having a third cross-sectional shape defined in a major plane of the constructed surface and a third cross-sectional area, where the sum of the first cross-sectional area and the second cross-sectional area is substantially equal to or less than the third cross-sectional area. For example, the example recess **380** is configured with a cross-sectional area that is smaller than the total cross sectional area of the tubular assembly **350** as shown in example configuration of FIG. 4A.

[0089] At **830**, the compliant jacket is at least partly conformed from the second cross-sectional shape to the third cross-sectional shape. In some implementations, the process **800** can include modifying the second cross-sectional shape to a fourth cross-sectional shape, different from the second cross-sectional shape, at least partly conforming the compliant jacket from the fourth cross-sectional shape to the third cross-sectional shape. For example, the example tubular assembly **350** is deformed from an initial shape (e.g., as shown in FIG. 4A) and inserted into the recess **380** (e.g., FIG. 4B), and is then compressed to partly flow and at least partly fill the recess **380** (e.g., FIG. 4C).

[0090] In some implementations, the process **800** can include modifying the second cross-sectional shape to a fourth cross-sectional shape, different from the second cross-sectional shape further comprises compressing the compliant jacket such that the second cross-sectional area to a fourth cross-sectional area less than the second cross-sectional area. In some implementations, at least partly conforming the compliant jacket from the fourth cross-sectional shape to the third cross-sectional shape can include expanding the compliant jacket from the fourth cross-sectional area to a fifth cross-sectional area larger than the fourth cross-sectional area. For example, the example tubular assembly **650** is deformed from the undistorted configuration **651** (e.g., as shown in FIG. 6A) to the compressed configuration **653**, is then inserted into the recess **680** (e.g., FIG. 6B), and is then allowed to partly re-expand to at least partly fill the recess **680** (e.g., FIG. 6C).

[0091] In some implementations, the compliant jacket can include a substantially incompressible material, and the fourth cross-sectional shape has substantially the second cross-sectional area. For example, the example protective jacket **554** of FIGS. 5A-5C has a cross-sectional shape that is substantially the same as the cross-sectional shape of the example recess **480**.

[0092] In some implementations, the tubular body can be arranged below the major plane. For example, in the example configurations shown in FIGS. 4A-6C, the example tubular bodies **352**, **552**, and **652** are arranged below the planes define by the corresponding top surfaces **302**, **502**, and **602**.

[0093] In some implementations, the process **800** can include modifying a portion of the constructed surface to form the recess below the major plane of the constructed surface. For example, as shown in FIGS. 2A, 2B, 3A, and 3B, the tools **270** and **370** are used to remove portions of the constructed surfaces **201** and **301** to define the recesses **280** and **380**.

[0094] In some implementations, at least partly conforming the compliant jacket from the second cross-sectional shape to the third cross-sectional shape can include applying a compressive force over the tubular assembly. For example, as shown in the examples of FIGS. 4A-4C, the example tubular assembly **350** can be pressed or “squished” into the recess **380** to cause the protective jacket **354** to conform to the shape of the recess **380**.

[0095] In some implementations, applying a compressive force over the tubular assembly can include rolling a wheel over the tubular assembly and the recess. Examples of devices that can be used to apply tubular members to recesses will be discussed in more detail in the descriptions of FIGS. 9-11.

[0096] FIG. 9 illustrates a perspective view of a tubular assembly deployment apparatus **900** according to embodiments of the disclosure. An example embodiment of a tubular assembly deployment apparatus is disclosed in U.S. Pat. No. 10,866,380, which is hereby incorporated by

reference herein in its entirety. The tubular assembly deployment apparatus **900** can include a spool **916** mounted to a forward end **905** of a chassis **926** of a main body **904**. The spool **916** can hold a predetermined length of tubular assembly **950** such as optical fiber or communication line (e.g., the tubular assemblies **150a**, **150b**, **250**, **350**, **550**, **650**). As illustrated, the tubular assembly **950** can be wound around the spool **916**. The tubular assembly **950** can unwind from spool **916** at a rate commensurate with the apparatus speed in the direction of travel **930** (e.g., advancing direction) across a constructed surface **901**. The tubular assembly **950** can pass through a tubular assembly applicator **914** before the tubular assembly **950** is placed directly on or in the respective layer. The tubular assembly applicator **914** can be mounted to the chassis **926**. The tubular assembly applicator **914** can be a ring, tube, or similar structure to guide the application of the tubular assembly **950** on or in the respective layer.

[0097] In some embodiments, the tubular assembly deployment apparatus **900** can include a smoothing apparatus **924** as shown in FIG. **9**. For example, the tool **270** or **370** can be configured to contact the constructed surface **901** to form recesses with bottoms that are substantially smooth (e.g., relatively smoother than the proximal portions of the top surface of the constructed surface **901**). In other embodiments, the smoothing apparatus **924** can be omitted. For example, an embodiment of a tubular assembly deployment apparatus without a smoothing apparatus is disclosed in U.S. Pat. No. 9,588,315, which is hereby incorporated by reference herein in its entirety. Other examples of a tubular assembly deployment apparatus are disclosed in International Patent Publication No. WO 2022/010553, which is also hereby incorporated by reference herein in its entirety.

[0098] After the tubular assembly **950** is deployed onto or at least partly into the top surface of the constructed surface **901**, a container and protectant applicator **952** stores and ejects an uncured protectant **954** over the tubular assembly **950**. A shaping tool **960** is advanced over the uncured protectant **954** to urge the uncured protectant **954** toward a predetermined cross-sectional shape **956**. A finisher **958** proximal a rear end **906** is configured to cure the uncured protectant **954** into substantially the predetermined cross-sectional shape **956**. The shaped, uncured protectant **956** is then cured into a substantially solid form (e.g., by drying, by exposure to sun or other ultraviolet light, by application of a hardening agent).

[0099] In some embodiments, a tubular assembly deployment apparatus **900** can be used to deploy the tubular assembly **950** onto the top surface of a layer of road or other constructed surface before a subsequent layer is applied on top. For example, the tubular assembly **950** can be deployed onto a sub-base layer by the tubular assembly deployment apparatus **900** before a base layer is laid.

[0100] In other embodiments, the tubular assembly **950** can be deployed onto the top surface of a layer of constructed surface by the tubular assembly deployment apparatus **900** and then pushed into (“smushed into”) that layer using, for example, a roller machine. For example, the tubular assembly **950** can be deployed onto the top surface of the constructed surface **901** and pressed into the constructed surface **901** when the apparatus **900** compacts the constructed surface **901**. In some embodiments, in order to push the tubular assembly **950** into a layer, the layer may be capable of receiving the tubular assembly **950**. For example, if the layer comprises asphalt or concrete, the asphalt or concrete may not have cured yet. For another example, if the layer includes gravel, the gravel may not be compacted so densely that it cannot receive the tubular assembly **950**.

Optionally, a subsequent layer can be laid on top of the layer that the tubular assembly has been pushed in to, for example, when the tubular assembly **950** is pushed into a sub-base layer and a base layer can be laid on top.

[0101] In addition to the embodiments discussed above, when repaving an existing road, the tubular assembly **950** can be deployed after the old road is milled and before the new base layer is laid. Further, when adding a subsequent base layer to an existing road, the tubular assembly **950** can be laid on top of the old base layer before the new base layer is laid or it can be laid on top of and pushed into the new base layer.

[0102] In some embodiments, a tubular assembly **950** can be deployed while performing a step of the road forming process. For example, the tubular assembly **950** can be deployed while laying or compacting a sub-grade layer, a sub-base layer, or base layers. For example, road-building equipment can include a tubular assembly deployment apparatus to deploy the tubular assembly **950** while performing other roadbuilding steps. For example, as shown in FIG. **10**, a roller machine **1000** can be equipped with the tubular assembly deployment apparatus **900** to deploy the tubular assembly **950** in front of the roller machine **1000** so that the roller machine **1000** deploys the tubular assembly **950** while applying its compacting force to roll, smooth, and/or set the layer. For example, the roller machine **1000** has a roller **1010** that can function as a compressor to urge the tubular assembly **950** into a recess defined in the constructed surface **901**. The roller machine **1000** can thereby apply the tubular assembly **950**, apply and/or push it into the layer, and compact the layer in one continuous process.

[0103] For another example, as shown in FIG. **11**, an asphalt paver machine **1100** can be equipped with a tubular assembly deployment apparatus **900** to deploy a tubular assembly **950** while laying asphalt in front of the asphalt paver machine **1100** so that the asphalt paver machine **1100** deploys the tubular assembly **950** and lays asphalt with the tubular assembly **950** and/or on top of the tubular assembly **950**. The asphalt paver machine **1100** merely provides an example of one type of machine that can form a layer of a paved surface. Any type of paver machine or material-depositing machine that is used to form any layer of a paved surface could be equipped, similar to the example shown in FIG. **11**, to include a tubular assembly deployment apparatus **900** to deploy the tubular assembly **950** simultaneously with, or in conjunction to, forming the layer.

[0104] Although the foregoing embodiments describe a tubular assembly deployment apparatus **900** attached to a roller machine **1000** and an asphalt paver machine **1100**, the disclosed subject matter is not so limited. Instead, as one of ordinary skill in the art will appreciate, the deployment system can be attached to any piece of equipment used for road building, including, for example, a dump truck configured to store and dump one of asphalt and concrete, a miller configured to destroy a layer of an existing road, a sweeper configured to remove debris from a first surface, or a piece of equipment including a nozzle configured to distribute tack coat onto a surface.

[0105] Although a few implementations have been described in detail above, other modifications are possible. For example, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A method for adhering a tubular assembly onto a constructed surface, the method comprising: removing a portion of the constructed surface to form a recess below a major plane of the constructed surface; arranging the tubular assembly partly within the recess; applying an uncured protectant; and curing the uncured protectant to protectively encase the tubular assembly.
2. The method of claim 1, further comprising shaping the uncured protectant before curing the uncured protectant such that the uncured protectant has a different cross-sectional shape after the shaping than prior to the shaping.
3. The method of claim 1, wherein: the tubular assembly is applied directly to the recess; and the uncured protectant is applied directly onto the tubular assembly.
4. The method according to claim 1, wherein the curing of the uncured protectant forms a cured protectant that protectively encases the tubular assembly at least partly within the recess, the cured protectant being an outermost layer protecting the tubular assembly at least partly within the recess.
5. The method of claim 1, wherein the tubular assembly has a predetermined thickness, and the

recess has a depth less than the predetermined thickness.

6. The method of claim 1, further comprising: applying the uncured protectant onto the tubular assembly; and curing the uncured protectant to cover the tubular assembly.

7. The method of claim 1, further comprising: applying the uncured protectant within the recess; and curing the uncured protectant to cover the tubular assembly.

8. The method of claim 1, wherein a first portion of the tubular assembly is arranged below the major plane and a second portion of the tubular assembly extends beyond the major plane, and the uncured protectant is cured to protectively encase the first portion within the recess and cover the second portion beyond the major plane.

9. The method of claim 1, wherein the tubular assembly comprises an optical conductor.

10. An apparatus for adhering a tubular assembly to a constructed surface, the apparatus comprising: a main body moveable in an advancing direction, the main body possessing a forward end and a rear end; a receiver configured to store a tubular assembly having a predetermined thickness, the receiver being mounted on the main body; a container configured to store an uncured protectant, the container being mounted on the main body; a surface modification apparatus connected to the main body, the surface modification apparatus being configured to contact the constructed surface at a contact point to form a recess below a major plane of the constructed surface when the apparatus moves in the advancing direction; a tubular assembly applicator configured to receive a portion of the tubular assembly from the receiver and configured to arrange the tubular assembly partly within the recess; and a protectant applicator connected to the container and configured to eject the uncured protectant wherein the uncured protectant is configured to cure after the uncured protectant is applied.

11. The apparatus of claim 10, wherein the surface modification apparatus is a grinder or a wheel.

12. The apparatus of claim 10, wherein the uncured protectant is a resin.

13. The apparatus of claim 10, wherein the uncured protectant is configured to be cured by application of a catalyst configured to initiate a curing process.

14. The apparatus of claim 10, wherein the surface modification apparatus is configured to form the recess to a depth less than the predetermined thickness below the major plane.

15. The apparatus of claim 10, wherein the protectant applicator is configured to eject the uncured protectant onto the tubular assembly, and the protectant applicator is positioned such that ejected uncured protectant is applied after the tubular assembly has been arranged by the tubular assembly applicator.

16. The apparatus of claim 10, wherein the protectant applicator is configured to eject the uncured protectant into the recess, and the protectant applicator is positioned such that ejected uncured protectant is applied after the tubular assembly has been arranged by the tubular assembly applicator.

17. The apparatus of claim 10, wherein the protectant applicator is configured to eject the uncured protectant into the recess, and the protectant applicator is positioned such that ejected uncured protectant is applied before the tubular assembly is arranged by the tubular assembly applicator.

18. The apparatus of claim 10, wherein the tubular assembly applicator is configured to arrange the tubular assembly partly within the recess such that a first portion of the tubular assembly is arranged below the major plane and a second portion of the tubular assembly extends beyond the major plane.

19. A method for adhering a tubular assembly onto a constructed surface, the method comprising: receiving the tubular assembly comprising: a tubular body having a predetermined first cross-sectional shape and a predetermined first cross-sectional area; and a compliant jacket coaxially surrounding the tubular body and having a predetermined second cross-sectional shape and a predetermined second cross-sectional area; arranging the tubular assembly partly within a recess having a third cross-sectional shape defined in a major plane of the constructed surface and a third cross-sectional area, wherein a sum of the predetermined first cross-sectional shape and the

predetermined second cross-sectional area is substantially equal to or less than the third cross-sectional area; and at least partly conforming the compliant jacket from the predetermined second cross-sectional shape to the third cross-sectional shape.

20. An apparatus for adhering a tubular assembly to a constructed surface, the apparatus comprising: a main body moveable in an advancing direction, the main body possessing a forward end and a rear end; a receiver configured to store a tubular assembly, the receiver being mounted on the main body and the tubular assembly comprising: a tubular assembly having a predetermined first cross-sectional shape and a predetermined first cross-sectional area; and a compliant jacket coaxially surrounding the tubular assembly and having a predetermined second cross-sectional shape and a predetermined second cross-sectional area; a tubular assembly applicator configured to receive a portion of the tubular assembly from the receiver and configured to arrange the tubular assembly at least partly within a recess having a third cross-sectional shape defined in a major plane of the constructed surface and a third cross-sectional area, wherein a sum of a first cross-sectional area of the predetermined first cross-sectional shape and second cross-sectional area of the predetermined second cross-sectional shape substantially equals the third cross-sectional area; and a compressor configured to apply a compressive force over the tubular assembly to at least partly conform the compliant jacket from the predetermined second cross-sectional shape to the third cross-sectional shape.
